

scGPT and scVI Citation-Use Review — Consolidated Results

Machine outcomes and two-reviewer primary-use validation

This document brings the latest frozen scGPT and scVI citation-use results into one place. Counts are at the citing-study-record level unless physical-paper deduplication is stated.

1. How to read these numbers

- Primary-use labels are mutually exclusive: Apply to biological data; Extend or develop; Benchmark or evaluate; Mention only; and Other executed use.
- Coverage incomplete, citation not located, and machine unresolved or no final are evidence or process states. They are not evidence that the method was not used.
- Every percentage states its denominator. Machine-composition percentages use classified records only; validation metrics use the human-reviewed subset.

2. Machine-review outcomes

Measure	scGPT	scVI
Analysis frame	1,291 frozen citation records	1,634 machine-reviewed evidence records
Machine-reviewed frame	726 rights-approved, parser-eligible records (56.24% of 1,291)	1,634 records in the latest harmonized aggregate
Classified primary use	402 / 726 (55.37%)	951 / 1,634 (58.20%)
Coverage incomplete	226 / 726 (31.13%)	536 / 1,634 (32.80%)
Citation not located	13 / 726 (1.79%)	79 / 1,634 (4.83%)
Machine unresolved or no final	85 / 726 (11.71%)	68 / 1,634 (4.16%)
Method executed among classified	117 / 402 (29.10%)	795 / 951 (83.60%)
Upstream excluded or not reviewed	565: 354 full text unavailable; 200 rights not approved; 11 parse/coverage/context gate	Not represented in this 1,634-record aggregate

The scVI rescue pass re-reviewed 122 records and produced 103 newly classified results relative to its v3 base. The harmonized scVI aggregate therefore contains 951 classified records.

3. Primary-use composition among classified records

Primary-use label	scGPT: n (% of 402)	scVI: n (% of 951)
Apply to biological data	20 (4.98%)	382 (40.17%)
Benchmark or evaluate	55 (13.68%)	341 (35.86%)
Extend or develop	41 (10.20%)	66 (6.94%)
Mention only	285 (70.90%)	156 (16.40%)
Other executed use	1 (0.25%)	6 (0.63%)
Total	402 (100%)	951 (100%)

4. Human validation of primary use

Both studies used two blinded human reviewers followed by a consensus step. Scoring treats the human consensus primary-use label as the reference standard and requires one primary label per scored paper.

For scGPT, 40 classification papers from a 50-paper formal sample were scored; 10 additional papers audited evidence and abstention states. Physical-paper deduplication gives a 400-study classified estimand and 719 eligible physical papers.

For scVI, the same 50/40 structure was used. After the rescue update, 39 of the original 40 classification papers remain machine-classified; HV-032 is now a no-final abstention.

Validation measure	scGPT	scVI, latest aggregate
Formal human sample	50 papers	50 papers
Classification-score set	40 papers	40 papers
Current machine-classified overlap	40 papers	39 papers; HV-032 is now no final
Exact primary-use accuracy	31 / 40 (77.50%)	34 / 39 (87.18%)
End-to-end accuracy, counting current abstention as incorrect	31 / 40 (77.50%)	34 / 40 (85.00%)

Validation measure	scGPT	scVI, latest aggregate
Macro precision, unweighted	69.44%	79.31%
Macro recall, unweighted	71.36%	77.50%
Macro F1, unweighted	65.10%	78.22%
Supplementary design-weighted accuracy (not precision/recall)	88.42%	Not valid for the updated 951-record frame

Per-class values below are unweighted one-vs-rest hard-label results. They are operating points, not full precision–recall curves.

Primary label	scGPT: precision / recall / F1	scVI: precision / recall / F1
Mention only	100.0% / 100.0% / 100.0%	89.5% / 100.0% / 94.4%
Apply to biological data	44.4% / 100.0% / 61.5%	88.9% / 80.0% / 84.2%
Benchmark or evaluate	100.0% / 45.5% / 62.5%	88.9% / 80.0% / 84.2%
Extend or develop	33.3% / 40.0% / 36.4%	50.0% / 50.0% / 50.0%

5. Interpretation and reporting limits

- The category distributions describe two different citing corpora. Differences should not be interpreted as causal evidence that one method penetrated routine biological discovery more deeply.
- The scGPT design-weighted primary-use accuracy is 88.42% for its original 400-study physically deduplicated classified frame. Its unweighted accuracy is shown for direct transparency.
- The previous scVI sampling weights targeted an older 881-paper classified frame and are not valid for the updated 951-paper aggregate because the 103 newly classified papers had zero probability of entering the old sample.
- The current scVI validation is therefore descriptive. A population estimate for all 951 classified records requires a supplementary human sample from the newly classified set.
- OTHER_EXECUTED_USE is not included in the class-specific validation table because the audited classification subsets contain no estimable human-positive class support.

6. Sources and provenance

Machine outputs: the scGPT final report summary (report fingerprint e5cc7b6e...1197a) and the scVI harmonized manifest plus review-record JSONL (SHA-256 74618813...cf52). Human validation: the frozen two-reviewer consensus and primary-use scoring artifacts for each

method. For scVI, the original 50-paper crosswalk was rejoined to the latest harmonized aggregate before recalculation.